

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files 2 files

Unemployment in India.csv(text/csv) - 47043 bytes, last modified: 6/7/2026 - 100% done
Unemployment_Rate_upto_11_2020.csv(text/csv) - 18274 bytes, last modified: 6/7/2026 - 100% done
 Saving Unemployment in India.csv to Unemployment in India (1).csv
 Saving Unemployment_Rate_upto_11_2020.csv to Unemployment_Rate_upto_11_2020 (1).csv

```
df = pd.read_csv("Unemployment_Rate_upto_11_2020 (1).csv")
df.head()
```

1 to 5 of 5 entries  

index	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Region.1	lon
0	Andhra Pradesh	31-01-2020	M	5.48	16635535	41.02	South	1
1	Andhra Pradesh	29-02-2020	M	5.83	16545652	40.9	South	1
2	Andhra Pradesh	31-03-2020	M	5.79	15881197	39.18	South	1

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

```
(267, 9)
```

```
df.columns
```

```
Index(['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)',
      'Estimated Employed', 'Estimated Labour Participation Rate (%)',
      'Region.1', 'longitude', 'latitude'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 267 entries, 0 to 266
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Region                                267 non-null    object
1   Date                                  267 non-null    object
2   Frequency                             267 non-null    object
3   Estimated Unemployment Rate (%)        267 non-null    float64
4   Estimated Employed                     267 non-null    int64
5   Estimated Labour Participation Rate (%) 267 non-null    float64
6   Region.1                              267 non-null    object
7   longitude                             267 non-null    float64
8   latitude                              267 non-null    float64
dtypes: float64(4), int64(1), object(4)
memory usage: 18.9+ KB
```

```
df.describe()
```

	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	longitude	latitude
count	267.000000	2.670000e+02	267.000000	267.000000	267.000000
mean	12.236929	1.396211e+07	41.681573	22.826048	80.532425
std	10.803283	1.336632e+07	7.845419	6.270731	5.831738
min	0.500000	1.175420e+05	16.770000	10.850500	71.192400
25%	4.845000	2.838930e+06	37.265000	18.112400	76.085600
50%	9.650000	9.732417e+06	40.390000	23.610200	79.019300
75%	16.755000	2.187869e+07	44.055000	27.278400	85.279900

```
df.isnull().sum()
```

	0
Region	0
Date	0
Frequency	0
Estimated Unemployment Rate (%)	0
Estimated Employed	0
Estimated Labour Participation Rate (%)	0
Region.1	0
longitude	0
latitude	0

```
dtype: int64
```

```
df = df.dropna()
```

```
df.columns
```

```
Index(['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)',
      'Estimated Employed', 'Estimated Labour Participation Rate (%)',
      'Region.1', 'longitude', 'latitude'],
      dtype='object')
```

```
df.columns = [
    'State',
    'Date',
    'Frequency',
    'Estimated Unemployment Rate',
    'Estimated Employed',
    'Estimated Labour Participation Rate',
    'Region',
    'longitude',
    'latitude'
]
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

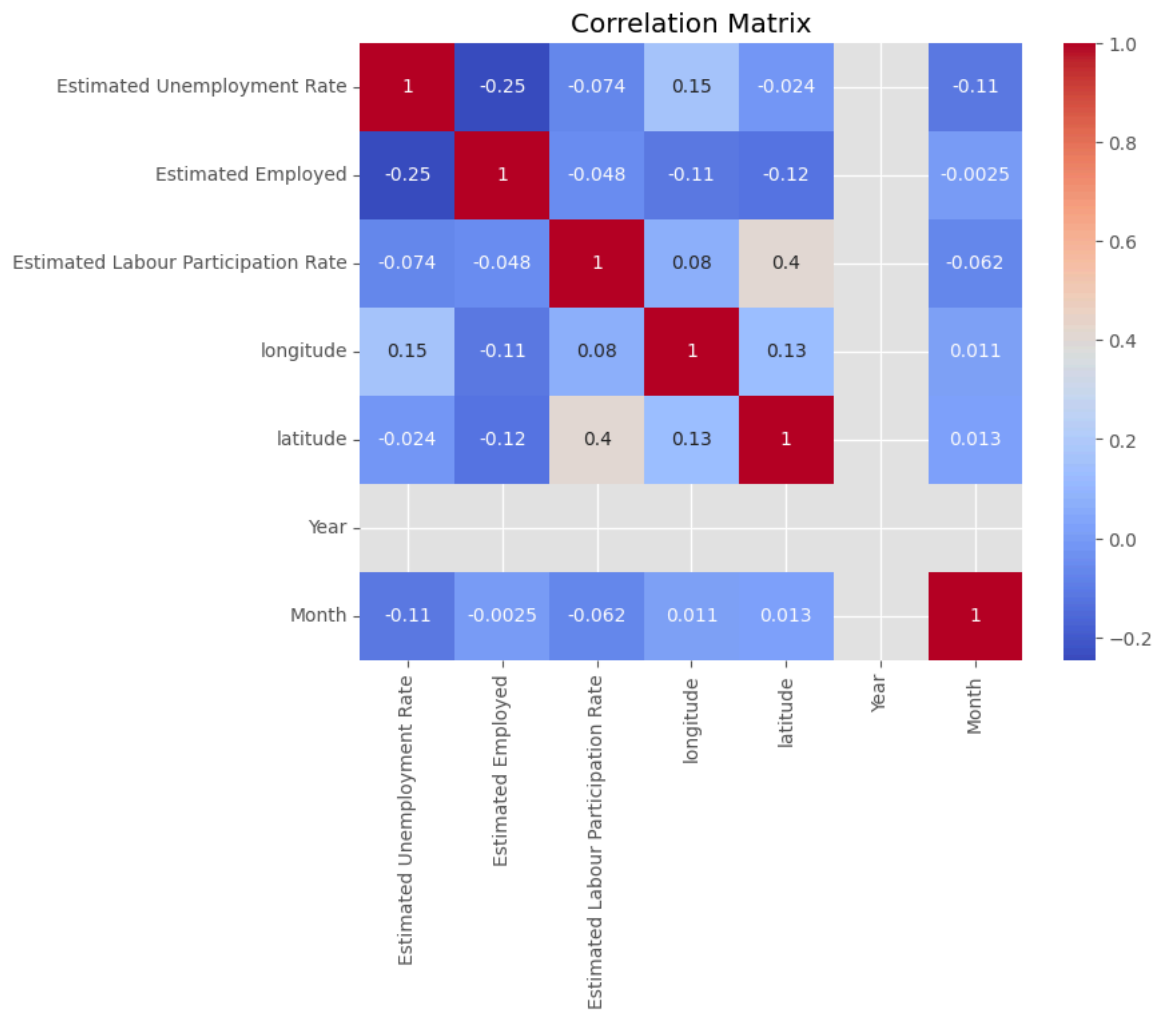
```
/tmp/ipykernel_4258/2394721818.py:1: UserWarning: Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was sp
df['Date'] = pd.to_datetime(df['Date'])
```

```
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
```

```
plt.figure(figsize=(8,6))

sns.heatmap(
    df.select_dtypes(include=np.number).corr(),
    annot=True,
    cmap='coolwarm'
)

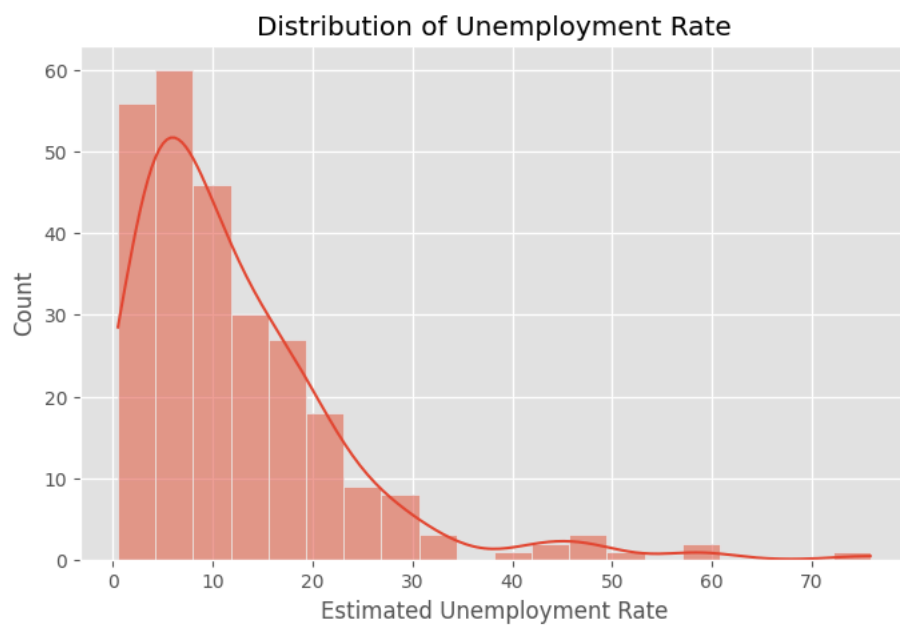
plt.title("Correlation Matrix")
plt.show()
```



```
plt.figure(figsize=(8,5))

sns.histplot(
    df['Estimated Unemployment Rate'],
    bins=20,
    kde=True
)

plt.title("Distribution of Unemployment Rate")
plt.show()
```



```
region_unemployment = df.groupby('Region')[
    'Estimated Unemployment Rate'
```

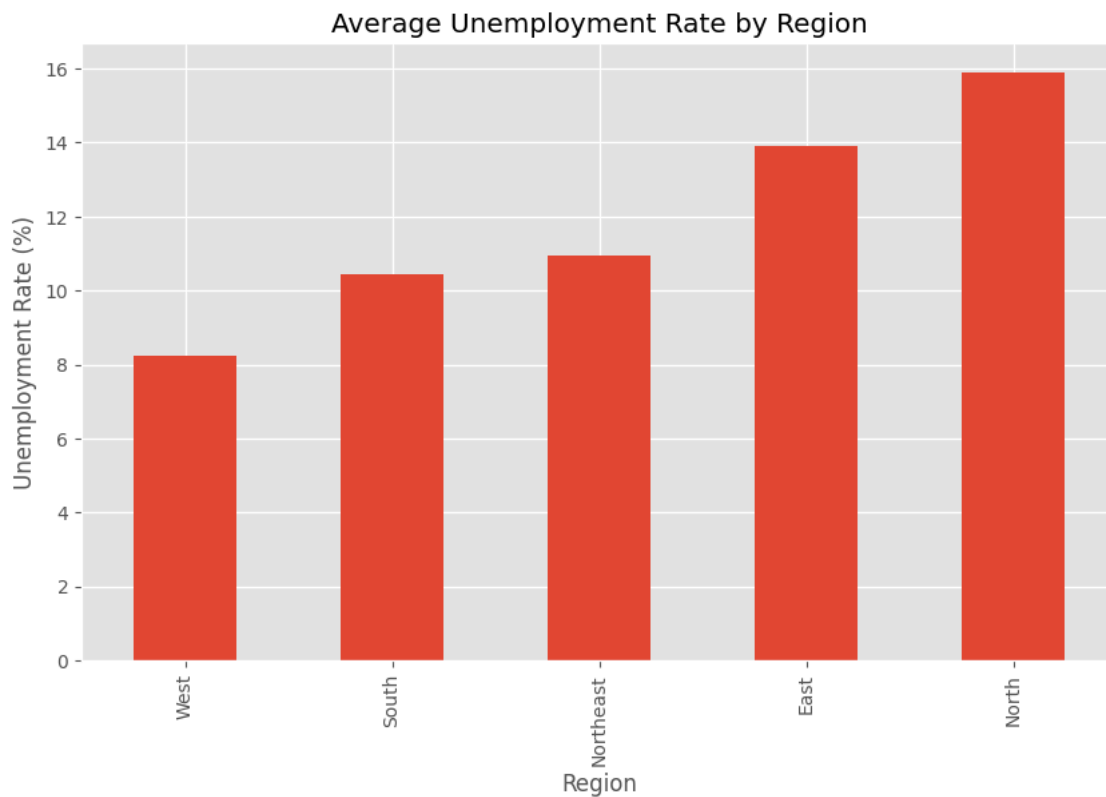
```
] .mean().sort_values()
```

```
plt.figure(figsize=(10,6))

region_unemployment.plot(kind='bar')

plt.title("Average Unemployment Rate by Region")
plt.ylabel("Unemployment Rate (%)")

plt.show()
```



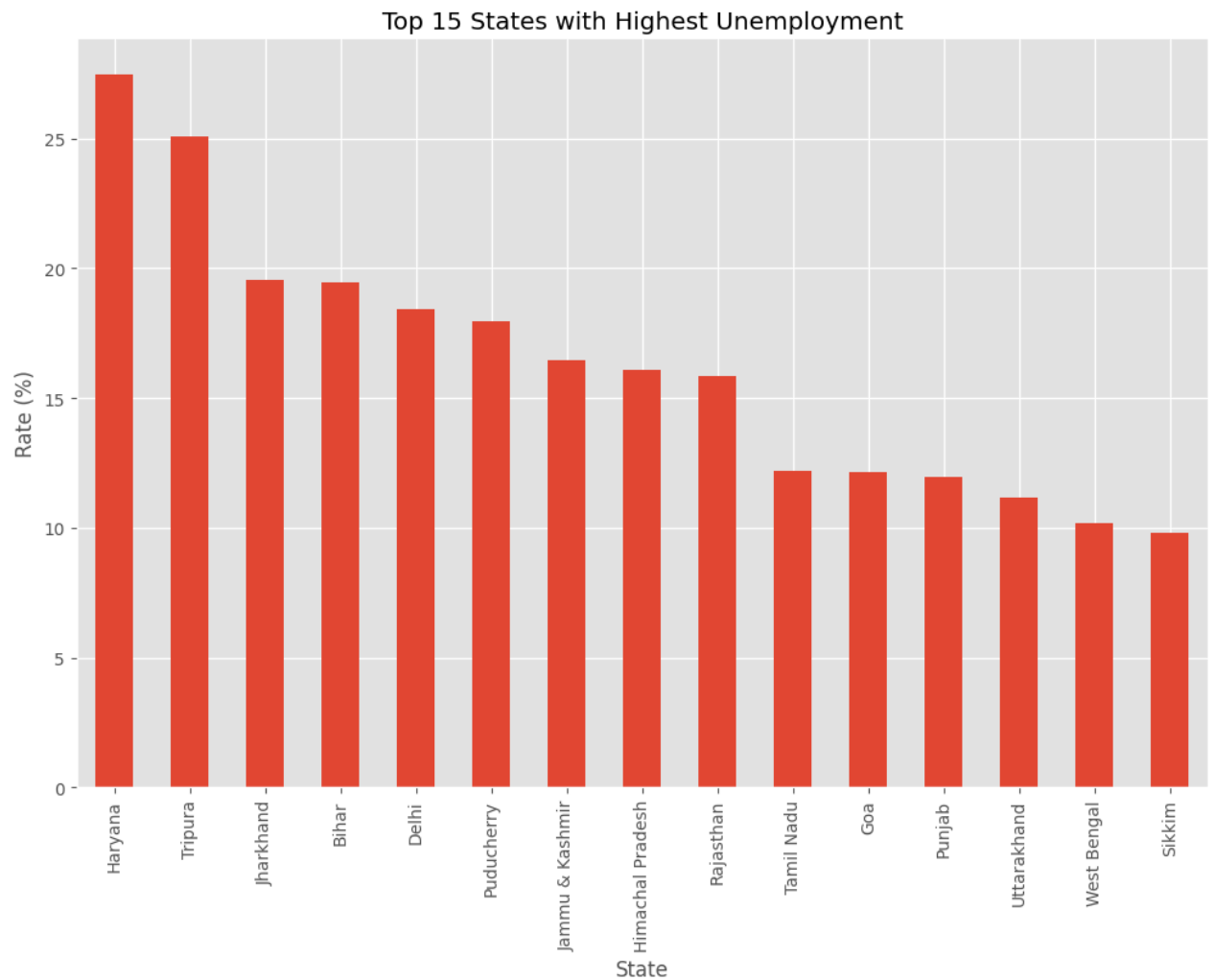
```
state_unemployment = df.groupby('State')[
    'Estimated Unemployment Rate'
].mean().sort_values(ascending=False)
```

```
plt.figure(figsize=(12,8))

state_unemployment.head(15).plot(kind='bar')

plt.title("Top 15 States with Highest Unemployment")
plt.ylabel("Rate (%)")

plt.show()
```



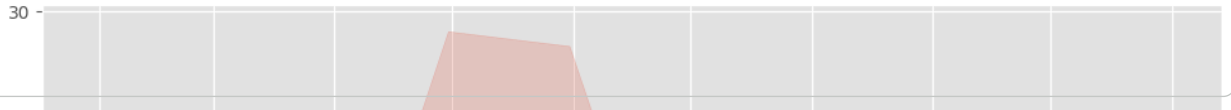
```
covid_period = df[df['Year'] >= 2020]
```

```
plt.figure(figsize=(12,6))

sns.lineplot(
    data=covid_period,
    x='Date',
    y='Estimated Unemployment Rate'
)

plt.title("Unemployment During COVID-19")
plt.show()
```

Unemployment During COVID-19



```
plt.figure(figsize=(12,6))

sns.barplot(
    data=covid_period,
    x='Region',
    y='Estimated Unemployment Rate'
)

plt.xticks(rotation=45)

plt.title("COVID Impact by Region")
plt.show()
```

